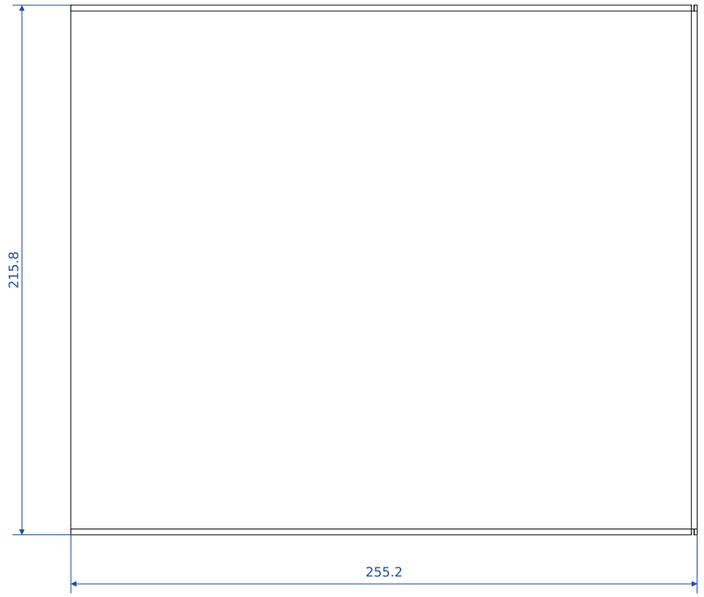
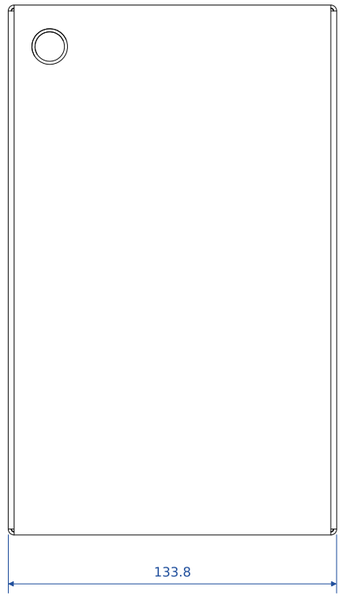
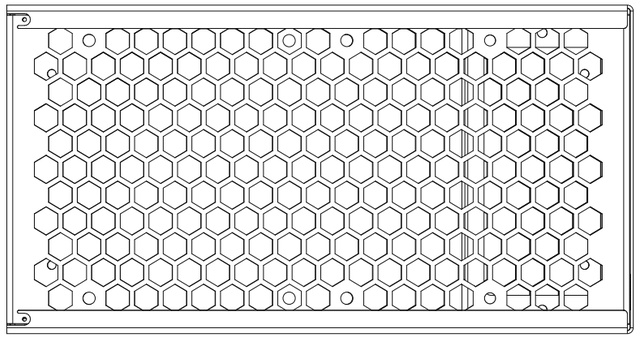
SIDE (open side; rear at left)



FRONT



TOP (from above; rear at left)



GENERAL NOTES

1. All dimensions in mm. The STEP model and the DXF flat patterns govern; do not scale this drawing.
2. Material: cold rolled low carbon steel, DC01 / SPCC / CR4 or equivalent, 1.2 thick.
3. Inside bend radius 1.2 nominal. Flat patterns are drawn for K = 0.44: adjust to your tooling so the formed sizes hold.
4. Tolerances unless stated: cut features +/-0.15, bend positions +/-0.4, angles +/-1 deg.
5. Deburr all edges and break sharp corners. Outside corners are R1.5 where drawn.
6. Self-clinching hardware: PennEngineering (PEM) or equivalent, installed before finishing.
7. Core, tray and bulkhead are joined with PEM SF-3-1.2 SpotFast fasteners, flush or under flush on both faces (sheet 2).
8. Finish: powder coat all over after joining, if offered; colour and texture to be agreed. Mask threads, studs and standoff bores.

PARTS (all 1.2 mm cold rolled steel)

no.	part	qty	mass	files
RMK1-01	Core: bottom, rear and top	1	709 g	core.step, core.dxf
RMK1-02	Mainboard tray	1	291 g	tray.step, tray.dxf
RMK1-03	Front bulkhead	1	244 g	bulkhead.step, bulkhead.dxf
RMK1-04	Power supply shelf	1	81 g	shelf.step, shelf.dxf
RMK1-05	Shell: front and sides	1	1370 g	shell.step, shell.dxf

HARDWARE AND JOINING BY THE FABRICATOR

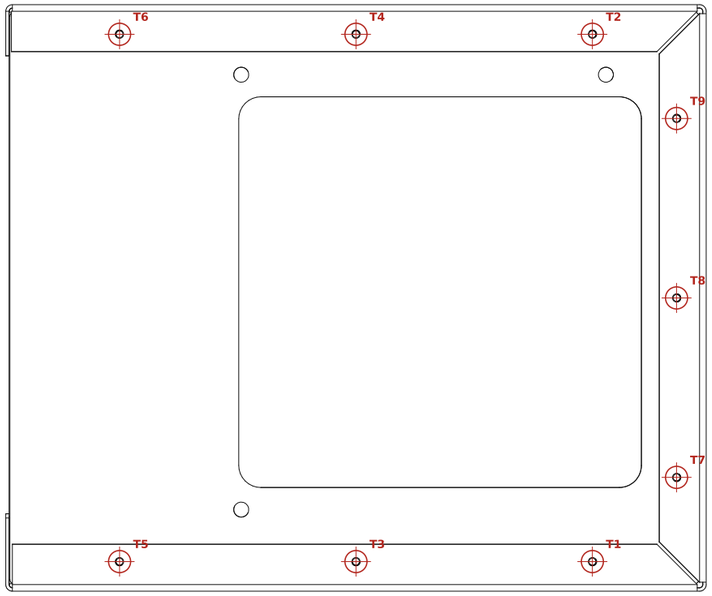
item		qty	where
PEM BSO-632-12	blind standoff, 6-32, 9.53 long	4	tray
PEM FH-632-4	flush-head stud, 6-32, 6.35 long	3	bulkhead
PEM FH-632-6	flush-head stud, 6-32, 9.53 long	4	core, bottom panel
PEM S-632-1	self-clinching nut, 6-32	2	shelf
PEM SF-3-1.2	SpotFast: tray to core 9, bulkhead to core 4	13	sheet 2
M2 x 0.4 tap	through, 1.6 drill	4	core

SCOPE

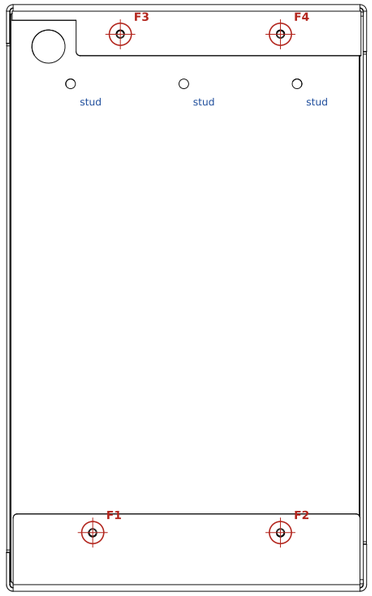
1. Five sheet metal parts. Core, tray and bulkhead are supplied joined as one assembly (sheet 2); shelf and shell are supplied loose.
2. Fans, mainboard, power supply, mains inlet, switch, screws and feet are fitted by the customer. The feet (sheet 8) are made separately.

TITLE	DRAWING NO.	SHEET / REV
Reactor Mk I - General arrangement	**RMK1-00**	1 of 8 rev A
MATERIAL	SCALE (A3)	DATE
see parts	1 : 2.5	2026-11-12

TRAY SIDE (front at left): 9 SpotFast, tray to the side lips of the core



FRONT: 4 SpotFast, bulkhead to front lips



SPOTFAST SF-3-1.2 (centres; origin below)

	X	Y	Z
T1	40.0	tray side	9.5
T2	40.0	tray side	200.5
T3	125.7	tray side	9.5
T4	125.7	tray side	200.5
T5	211.3	tray side	9.5
T6	211.3	tray side	200.5
T7	9.5	tray side	40.0
T8	9.5	tray side	105.0
T9	9.5	tray side	170.0
F1	front	30.0	20.0
F2	front	98.0	20.0
F3	front	40.0	200.5
F4	front	98.0	200.5

JOINING

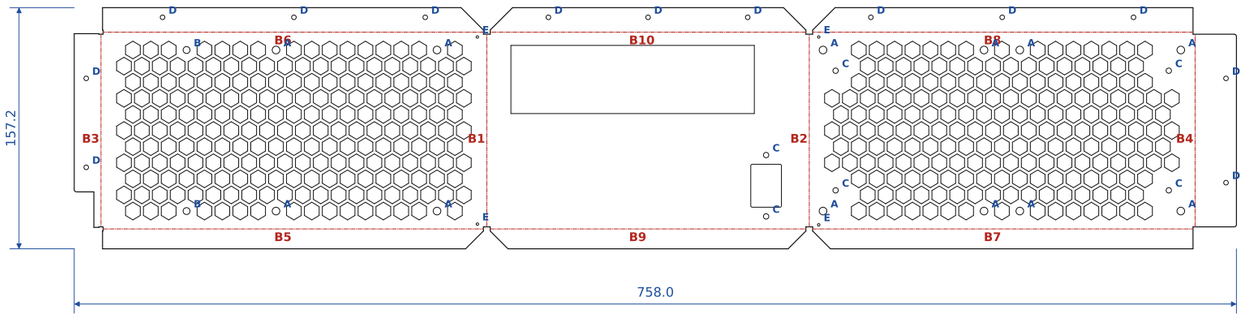
1. Datum: X from the inside face of the rear panel, Y from the inside face of the open-side lips, Z from the top face of the bottom panel.
2. Tray: its outer face against the inside of the 17 mm lips of the bottom, top and rear panels, 1.5 clear of each panel.
3. Bulkhead: its front face against the inside of the front lips of the top and bottom panels, 1.5 clear of each panel, its edge 0.3 from the tray.
4. Every fastener is 9.5 from the panel its lip is folded from, except F1 and F2 at Z 20. Lip flats are 14.6 or more.
5. Install the studs and standoffs first. Join the tray first, then the bulkhead.
6. SpotFast: 3.0 +0.08/0 holes in the core's lips, 2.5 in the tray and bulkhead, heads in the lips. Both faces must end flush: there is 0.3 to 0.5 outside the lips and 2.8 inside the tray.
7. Spot welding at the same centres is an accepted alternative: nugget 5 min, upset dressed flush outside. Leave the 13 holes out.
8. After joining the open side must stay square: 128.0 +/-0.4 between the side lips, 210.0 +/-0.4 between top and bottom panels.

NOTES

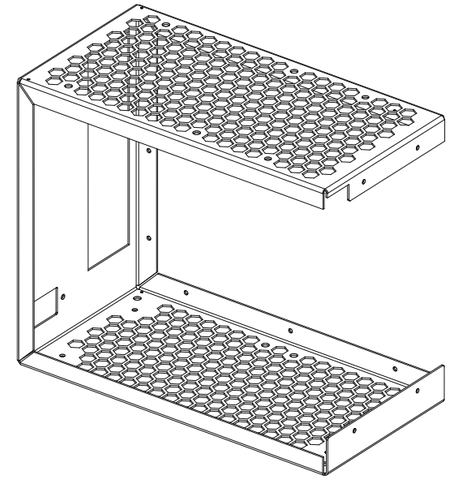
1. Outside faces of the lips are under the shell, which slides over them 0.5 clear: nothing may stand proud of them.
2. Studs marked in the front view are FH-632-4 in the bulkhead, heads flush at the front.

TITLE **Reactor Mk I - Joined assembly: core, tray, bulkhead**	DRAWING NO. **RMK1-10**	SHEET / REV 2 of 8 rev A
MATERIAL RMK1-01, -02, -03	SCALE (A3) 1 : 2.22	DATE 2026-11-12

FLAT PATTERN (bend lines dash-dot, numbered as in core-bends.csv)



FORMED (not to scale)



PART

Formed size	253.7 x 130.4 x 212.4
Flat size	758.0 x 157.2
Bends	10, all 90 deg, inside R1.2
Mass	709 g

ROUND HOLES

	dia	qty	for
A	5	12	fan screw, self-tapping, clearance
B	4.5	2	shelf screws 6-32, loose clearance
C	3.5	6	+0.08/0: PEM FH-632-6 stud, bottom panel (4); mains inlet M3, rear panel (2)
D	3	13	PEM SF-3-1.2 SpotFast, head side, +0.08/0
E	1.6	4	tap M2 x 0.4 through

BENDS

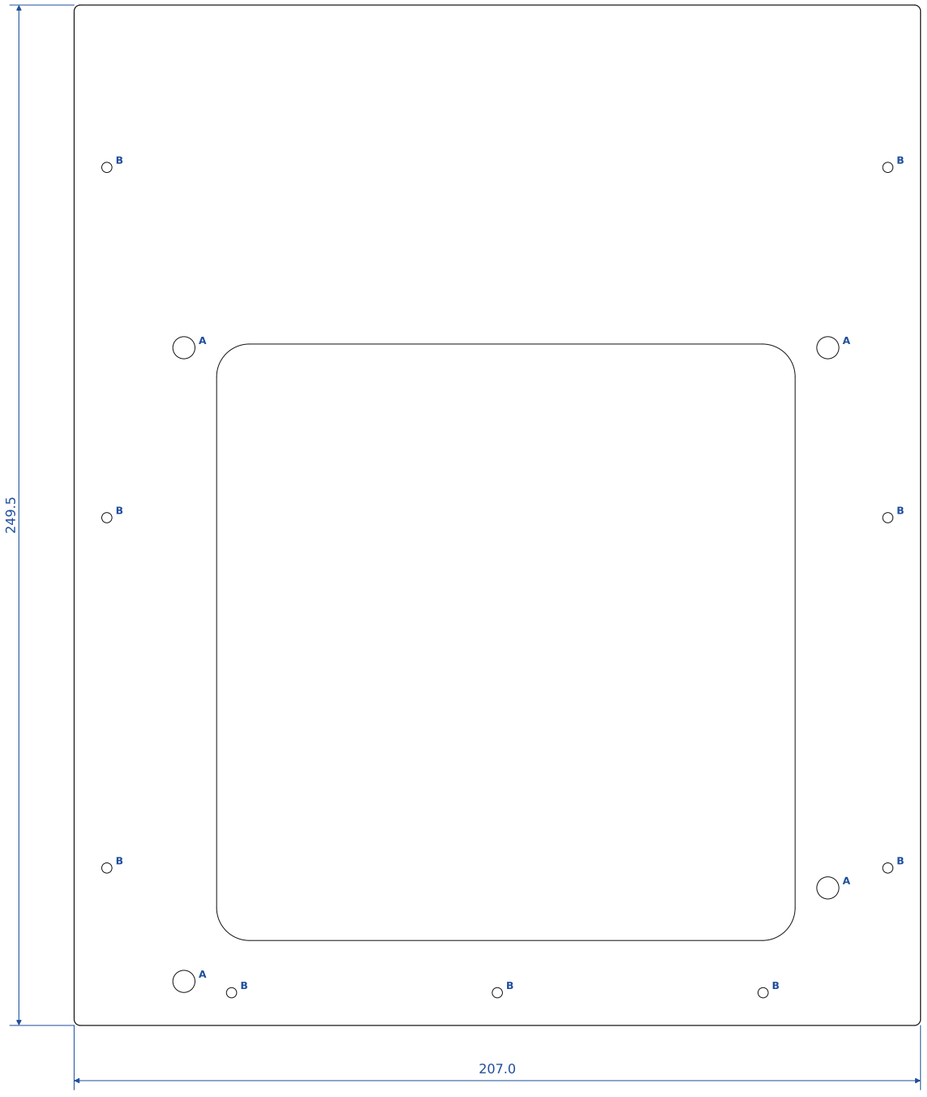
	angle	dir.	length
B1	90	up	125.6
B2	90	up	125.6
B3	90	up	125.6
B4	90	up	125.6
B5	90	up	248.1
B6	90	up	248.1
B7	90	up	248.1
B8	90	up	248.1
B9	90	up	205.9
B10	90	up	205.9

NOTES

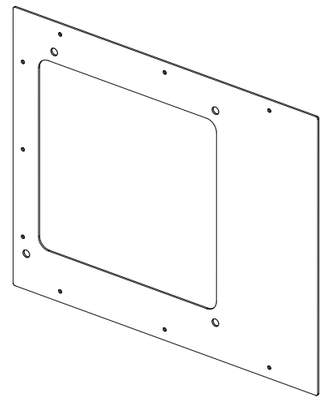
1. Hexagonal mesh: 9.6 across flats, webs 2.07; see DXF.
2. Rear panel: I/O aperture 158.75 x 44.45 +/-0.2 to ATX 2.2 fig. 4; keep paint off both faces within 2.5 of it.
3. Mains inlet cut-out 28 x 20 +0.2/0, R1, for Schurter 6100-3 mounted from outside.
4. M2 threads (4) at the rear corners of the top and bottom panels: tap after bending.
5. Four PEM FH-632-6 studs in the bottom panel for the feet, heads flush in its upper (inside) face, pointing down.
6. General notes on sheet 1 apply.

TITLE **Reactor Mk I - Core: bottom, rear and top**	DRAWING NO. **RMK1-01**	SHEET / REV 3 of 8 rev A
MATERIAL CRS DC01, 1.2 mm	SCALE (A3) flat 1 : 4	DATE 2026-11-12

FLAT PATTERN (bend lines dash-dot, numbered as in tray-bends.csv)



FORMED (not to scale)



PART

Formed size	249.5 x 1.2 x 207.0
Flat size	207.0 x 249.5
Bends	none
Mass	291 g

ROUND HOLES

	dia	qty	for
A	5.41	4	PEM BSO-632-12 standoff, +0.08/0
B	2.5	9	PEM SF-3-1.2 SpotFast, second panel: see PEM bulletin SF

NOTES

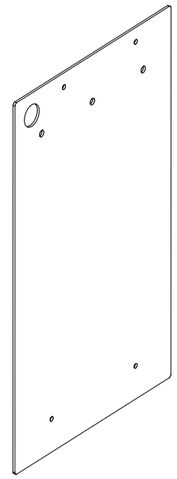
1. Flat. Four PEM BSO-632-12 standoffs pressed in from the face that goes against the core's lips, standing 9.53 off the other face.
2. Install the standoffs before joining: two heads end up under the core's lips.
3. General notes on sheet 1 apply.

TITLE	DRAWING NO.	SHEET / REV
Reactor Mk I - Mainboard tray	**RMK1-02**	4 of 8 rev A
MATERIAL	SCALE (A3)	DATE
CRS DC01, 1.2 mm	flat 1 : 1.5	2026-11-12

FLAT PATTERN (bend lines dash-dot, numbered as in bulkhead-bends.csv)



FORMED (not to scale)



PART

Formed size	1.2 x 126.1 x 207.0
Flat size	207.0 x 126.1
Bends	none
Mass	244 g

ROUND HOLES

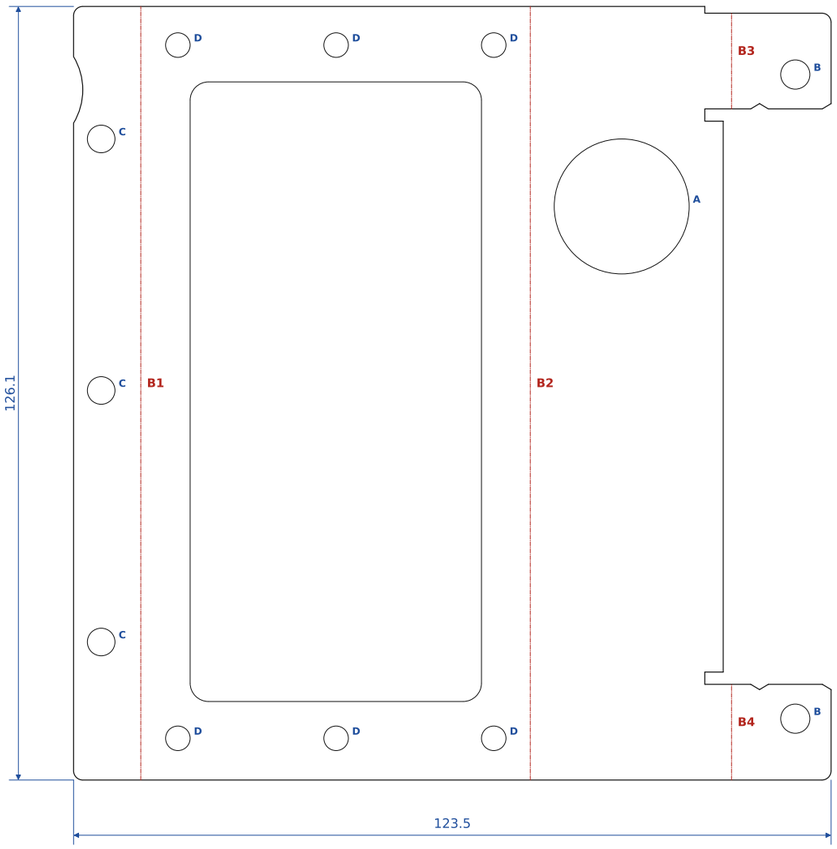
	dia	qty	for
A	12	1	power switch, +0.2/0
B	3.5	3	PEM FH-632-4 stud, +0.08/0
C	2.5	4	PEM SF-3-1.2 SpotFast, second panel: see PEM bulletin SF

NOTES

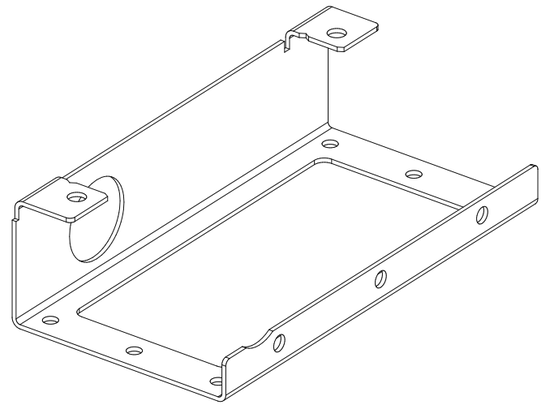
1. Flat. Three PEM FH-632-4 studs, heads flush in the front (outside) face, pointing into the case.
2. General notes on sheet 1 apply.

TITLE **Reactor Mk I - Front bulkhead**	DRAWING NO. **RMK1-03**	SHEET / REV 5 of 8 rev A
MATERIAL CRS DC01, 1.2 mm	SCALE (A3) flat 1 : 1	DATE 2026-11-12

FLAT PATTERN (bend lines dash-dot, numbered as in shelf-bends.csv)



FORMED (not to scale)



PART

Formed size	65.6 x 126.1 x 34.9
Flat size	123.5 x 126.1
Bends	4, all 90 deg, inside R1.2
Mass	81 g

ROUND HOLES

	dia	qty	for
A	22	1	cable grommet
B	4.76	2	PEM S-632-1 nut, +0.08/0
C	4.5	3	bulkhead studs, loose clearance
D	4	6	6-32 clearance, supply: SFX pattern both ways round

BENDS

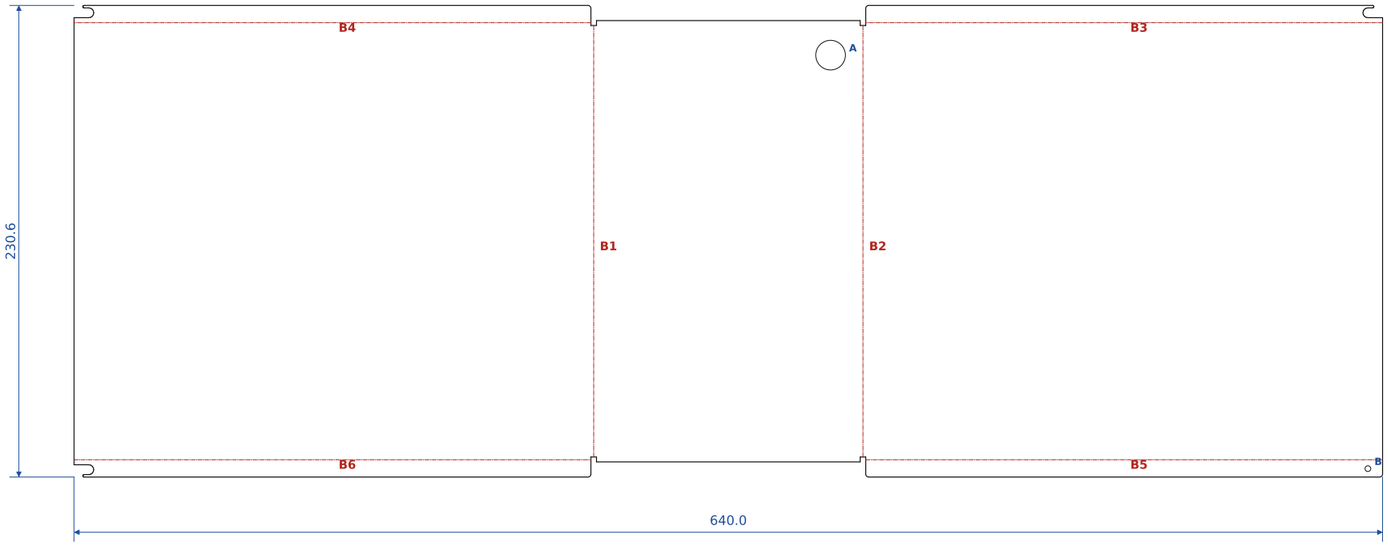
	angle	dir.	length
B1	90	up	126.1
B2	90	up	126.1
B3	90	up	15.6
B4	90	up	15.6

NOTES

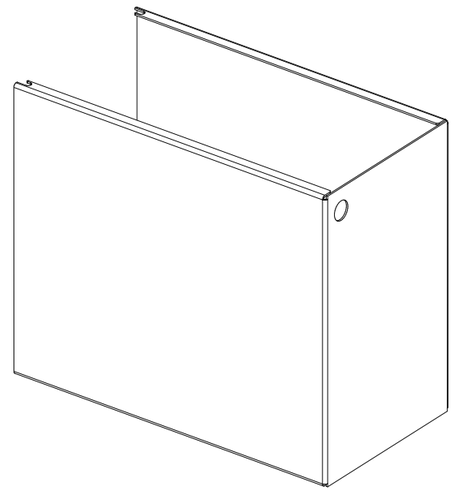
1. Two PEM S-632-1 nuts in the tabs, bodies under the tabs, flush on top.
2. Supplied loose: it hangs on the bulkhead's studs and two screws through the top panel.
3. General notes on sheet 1 apply.

TITLE **Reactor Mk I - Power supply shelf**	DRAWING NO. **RMK1-04**	SHEET / REV 6 of 8 rev A
MATERIAL CRS DC01, 1.2 mm	SCALE (A3) flat 1 : 1	DATE 2026-11-12

FLAT PATTERN (bend lines dash-dot, numbered as in shell-bends.csv)



FORMED (not to scale)



PART

Formed size	255.2 x 133.8 x 215.8
Flat size	640.0 x 230.6
Bends	6, all 90 deg, inside R1.2
Mass	1370 g

ROUND HOLES

	dia	qty	for
A	14.5	1	power switch head, 14
B	2.8	1	M2 clearance

BENDS

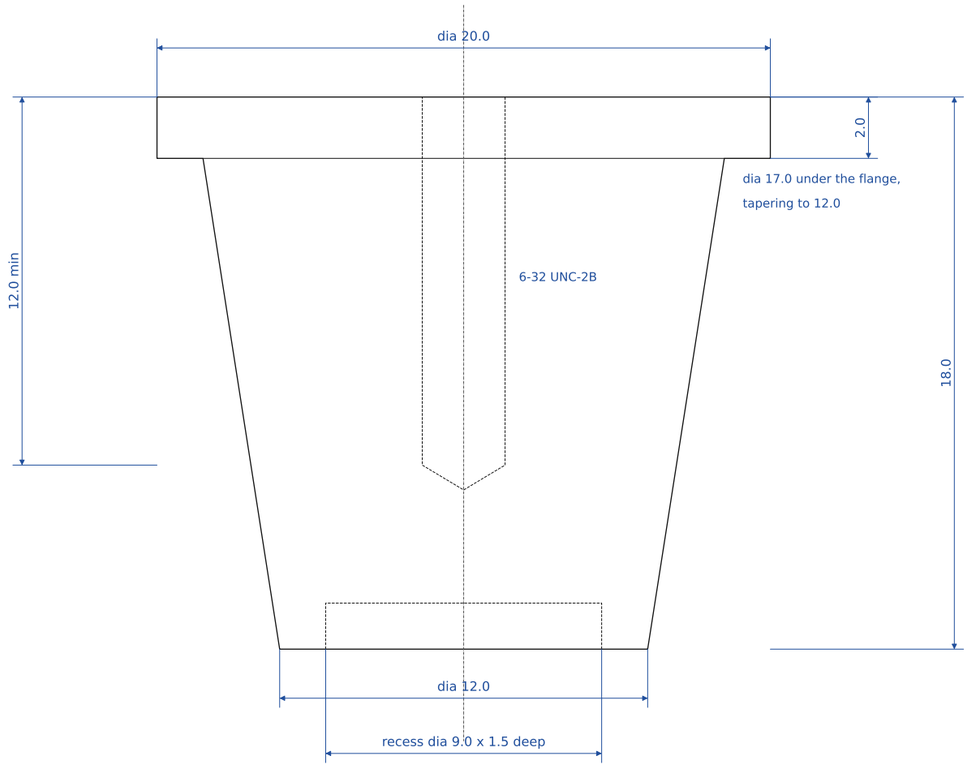
	angle	dir.	length
B1	90	up	211.0
B2	90	up	211.0
B3	90	up	252.8
B4	90	up	252.8
B5	90	up	252.8
B6	90	up	252.8

NOTES

1. Plain outside faces: no marks, grain or tooling witness on the front and sides.
2. Slots in three return flanges run onto M2 screw heads on the core; check the slide fit before finishing.
3. General notes on sheet 1 apply.

TITLE	DRAWING NO.	SHEET / REV
Reactor Mk I - Shell: front and sides	**RMK1-05**	7 of 8 rev A
MATERIAL	SCALE (A3)	DATE
CRS DC01, 1.2 mm	flat 1 : 3	2026-11-12

SIDE (flange up, against the bottom panel; hole and recess shown hidden)



PART

Quantity	4
Model	foot.step
Mass	about 25 g
Thread	6-32 UNC-2B, 2.7 drill, 12 deep

NOTES

1. Printed in 316L stainless steel by laser powder bed fusion or binder jetting.
2. Tap 6-32 UNC-2B after printing, 10 of full thread or more. foot.step has the 2.7 tap drill hole, not the thread; drill it to size first if it prints small.
3. The flange's top face sits against the case: flat within 0.1, no support marks.
4. Recess underneath for a stick-on bumper, 3M Bumpon SJ5076 (8.0 dia x 2.8), which stands 1.3 below the rim.
5. Each foot turns onto a PEM FH-632-6 stud standing 9.5 out of the bottom panel, hand tight with thread lock.
6. Finish to match the case.

TITLE	DRAWING NO.	SHEET / REV
Reactor Mk I - Foot	**RMK1-06**	8 of 8 rev A
MATERIAL	SCALE (A3)	DATE
316L stainless steel, printed	5 : 1	2026-11-12